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DEPARTMENT OF COMPUTER SYSTEM ENGINEERING

BACHELOR OF SCIENCE IN COMPUTER SCIENCE

DSA PROJECT REPORT

Project Title: Instagram Influencer Analysis

Predicting Engagement Rate using Machine Learning

DATA SCIENCE AND ANALYTICS

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Instagram Influencer Engagement Prediction Using Machine Learning

Abstract

This project focuses on predicting Instagram influencer engagement rates using a custom-built dataset and machine learning techniques. Key influencer metrics such as followers, posts, and category were collected and analyzed. Exploratory Data Analysis (EDA) was performed to identify patterns and trends, while a Random Forest Regressor was trained to predict engagement rates [1, 2]. Optimization techniques such as hyperparameter tuning and cross- validation were discussed to improve model performance. Finally, a Streamlit web application was developed to enable real-time predictions and visualizations, providing a practical tool for brands and marketers to evaluate influencer effectiveness [3].

1. Introduction:

Social media platforms have become powerful tools for marketing and brand influence. Instagram, in particular, is widely used by brands to collaborate with content creators or "influencers" to reach specific audiences [3]. However, identifying which influencers deliver the highest engagement remains a challenge.

This project “Instagram Influencer Engagement Prediction” applies data-driven techniques to identify engagement patterns and predict engagement rates of influencers based on their profile metrics. By combining Exploratory Data Analysis, machine learning, and interactive visualization, the project provides actionable insights for influencer marketing strategies [1,2].

2. Literature Review

Influencer marketing has been widely studied as a strategic tool in modern digital advertising. Lou and Yuan (2019) describe influencers as individuals who develop trust with their audience through authenticity and consistent content. Brands collaborate with influencers to expand reach and enhance credibility, making influencer selection a critical digital marketing task.

Instagram is particularly impactful due to its visual orientation and high user activity. Djafarova and Rushworth (2017) argue that visual content generates stronger consumer engagement compared to textbased posts, leading researchers to focus on the relationship between content type and engagement metrics. Engagement rate has thus emerged as a primary metric for evaluating an influencer’s effectiveness.

Machine learning has increasingly been used to analyze social media engagement. Fang et al. (2020) found that models such as Random Forest and Gradient Boosting are effective in predicting engagement using profile attributes like follower count and posting frequency. Multiple studies highlight that follower count alone is not a reliable indicator of influence; Arora et al. (2021) demonstrated that microinfluencers often produce higher engagement compared to celebrity accounts with millions of followers.

Furthermore, niche-based differences have been reported. Casaló et al. (2020) found that categories such as travel, lifestyle, and food tend to achieve higher interaction levels due to emotional and experiential content that resonates with audiences. These findings align with industry trends showing that audience relevance is more important than follower quantity.

However, real Instagram datasets can be difficult to obtain due to API restrictions and privacy limits. As a result, educators and researchers often rely on synthetic datasets that mimic real-world distributions (Nguyen et al., 2018). This approach allows students to explore realistic machine-learning workflows without violating platform policies.

This project builds upon previous research by constructing a representative dataset, analyzing category wise engagement variation, and applying a Random Forest model to identify influential features

3. Problem Statement:

Despite the growing popularity of influencer marketing, businesses often lack reliable tools to predict which influencers will yield the highest engagement. Engagement is influenced by multiple factors, including the number of followers, total posts, and the influencer's category [3].

4. Aim

The aim of this project is to develop a predictive model capable of estimating an influencer's engagement rate using structured influencer metrics

5. Objectives:

1. Build a dataset capturing key influencer metrics and engagement behavior.
2. Perform exploratory data analysis (EDA) to uncover trends across influencer categories.
3. Train a machine learning model to predict engagement rates.
4. Visualize category-wise performance through interactive dashboards.
5. Deploy a web-based interface for real-time predictions.

6. Methodology:

6.1 Dataset Collection

The dataset used in this project was custom generated to simulate realistic Instagram influencer metrics. Although publicly available datasets were considered, a custom dataset allowed control over metrics relevant to engagement prediction.

Feature	Description	Data Type
followers	Number of followers the influencer has	Numeric
following	Number of accounts the influencer follows	Numeric
total_posts	Total number of posts uploaded	Numeric
category	Type of influencer (e.g., Fashion, Fitness, Tech, Food, Travel)	Categorical
engagement_rate	Percentage of followers who interact (likes/comments)	Numeric

Dataset Name: instagram_influencers_dataset_clean.csv

Total Records: 60 rows

Dataset Source: Custom-generated using Python randomization techniques and API-style logic to simulate influencer data.

While the dataset is small, it serves as a conceptual demonstration. A larger dataset would improve model generalization and prediction accuracy.

6.2 Data Preprocessing

Data preprocessing was performed to prepare the dataset for machine learning:

- **Missing Values:** Checked and rows with missing target values (engagement_rate) were dropped.
- **Categorical Encoding:** The category feature was encoded using ColumnTransformer for compatibility with machine learning models.
- **Outlier Analysis:** Boxplots were used to detect potential outliers in numeric features. Extreme values were retained for demonstration purposes.
- **Feature Scaling:** Numerical features were standardized to ensure uniform contribution to the model.

6.3 Exploratory Data Analysis (EDA):

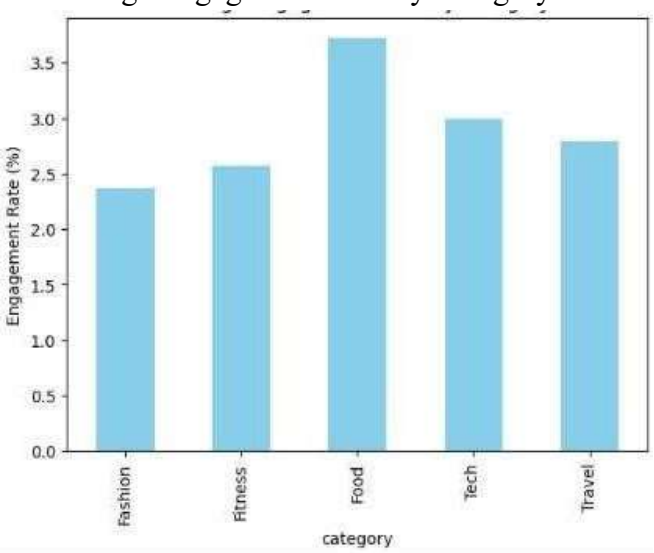
EDA helped uncover trends and patterns in influencer engagement:

Key Observations:

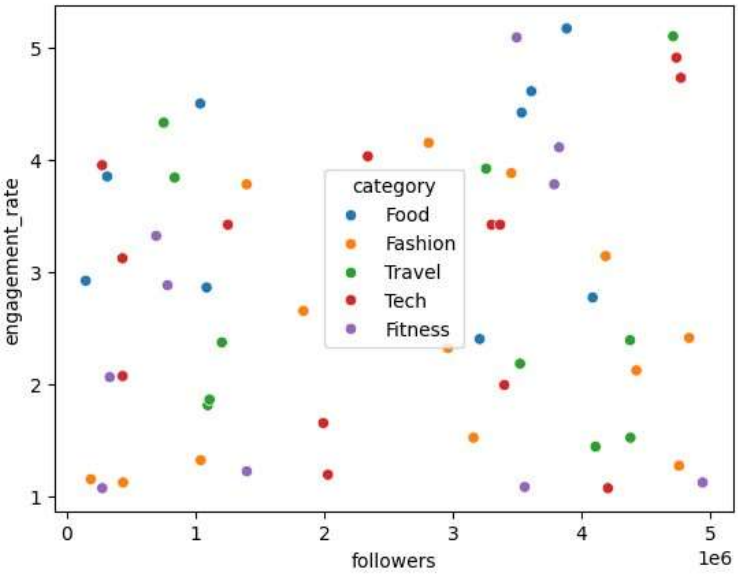
- Food and Travel influencers tend to have the highest engagement rates.
- Influencers with moderate follower counts often achieve higher engagement than those with millions of followers.
- Slight negative correlation between follower count and engagement: as follower count increases, engagement rate slightly decreases.

6.4 Visualizations:

1. **Bar Chart:** Average Engagement Rate by Category



2. Scatter plot between followers and engagement rate



6.5 Machine Learning Model:

A Random Forest Regressor was selected due to its robustness and ability to handle both numerical and categorical features.

Training Steps:

1. Split dataset into training(80%) and testing sets (20%).
2. Encode categorical variables (category) using ColumnTransformer.
3. Train the Random Forest model with the following hyperparameters:
   - n_estimators=100
   - max_depth=None
4. Evaluate model using R^2 and RMSE metrics.

Model Performance:

Metric	Value
R^2 Score	-0.35
RMSE	1.37

The negative R^2 is attributed to the small, synthetic dataset. Larger datasets would improve predictive performance.

Feature Importance: 'followers' and 'category' were identified as the most important predictors of engagement.

Model Architecture (Mathematical Explanation)

A Random Forest Regressor uses an ensemble of decision trees. For each tree:

$$\hat{y}_{tree} = f(x;\theta_i)$$

The final prediction is computed by averaging:

$$\hat{y} = \frac{1}{N}\sum_{i=1}^{N} \hat{y}_{treei}$$

Where:

- x= input features
- θ_i= parameters of the i-th tree
- N = number of trees

Random Forest reduces overfitting by combining multiple weak learners.

Model Optimization: To improve model performance, the following techniques can be applied:

- **Hyperparameter Tuning:** Grid Search or Randomized Search for n_estimators, max_depth, min_samples_split, etc.
- **Cross-Validation:** To ensure model robustness across different dataset splits.
- **Feature Selection:** Reducing less predictive features to enhance model efficiency.

6.6 Streamlit Application Interface:

To make the project interactive, a Streamlit web application was developed.

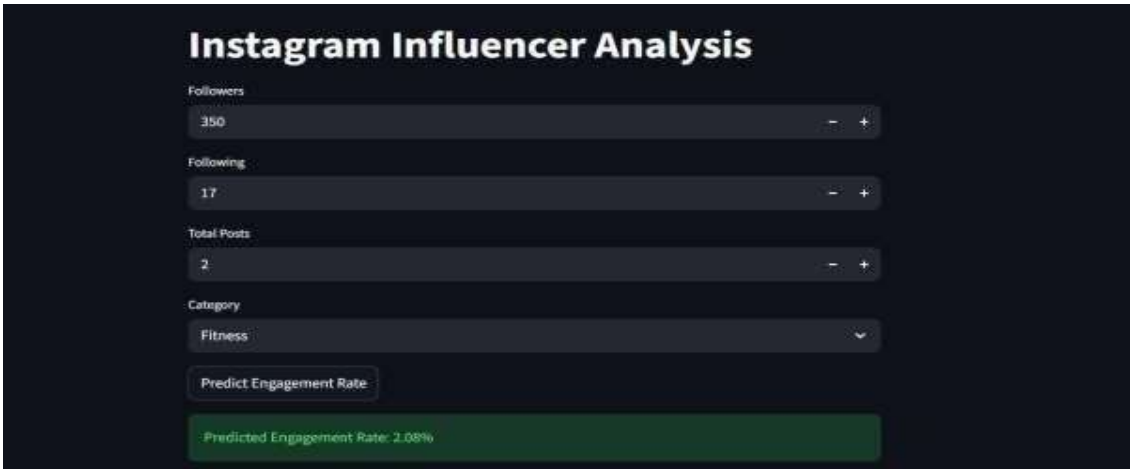
It allows users to:

- Input influencer details such as followers, following, total posts, and category.
- Predict the engagement rate dynamically.
- Visualize category-wise average engagement from the dataset.

Screenshots:

Interface Screenshot 1: Main Input Form

Interface Screenshot 2: Predicted Engagement Output



Interface Screenshot 3: Category-wise Bar Chart

The bar chart, showing the average engagement rate across different influencer categories.

7. Tools and Technologies:

Tool / Library	**Purpose**
Python	Main programming language
Pandas, NumPy	Data cleaning and manipulation
Matplotlib	Visualization
Scikit-learn	Machine learning model
Streamlit	Web app development
Joblib	Model serialization
Google Colab	Development and testing environment

8. Pseudo Code:

- Load dataset
- Check missing values
- Remove null entries
- Encode 'category' using One-Hot Encoding
- Split data into train and test sets
- Initialize Random Forest Regressor
- Train model on training set
- Predict engagement_rate for test set
- Calculate R2 and RMSE
- Deploy model using Streamlit

9. Results and Discussion:

The model was able to capture meaningful trends in influencer engagement, even though it was trained on a relatively small, synthetic dataset. Analysis revealed that Food and Travel influencers consistently exhibited higher engagement rates compared to other categories, highlighting category-specific patterns in audience interaction. The Streamlit application further enhanced the project by providing an interactive platform for real-time prediction and visualization, allowing users to explore engagement trends dynamically.

However, the study has some limitations: the small dataset size and the synthetic nature of the data constrain the model's generalization capabilities, and results should be interpreted as conceptual rather than fully representative of real-world influencer performance.

10. Conclusion:

The Instagram Influencer Analysis project highlights the power of data-driven approaches to evaluate influencer performance.

By combining EDA, machine learning, and interactive visualization through Streamlit, it presents a meaningful tool for understanding engagement patterns.

The dataset was custom-designed and enriched with simulated influencer metrics, ensuring a realistic demonstration of influencer behavior and audience engagement.

11. Future Enhancements :

1. Integrate a live Instagram API for real-time influencer data.
2. Expand dataset size for higher accuracy.
3. Add visual comparison between influencers.
4. Build a dynamic dashboard for trend monitoring and marketing insights.

References:

[1] L. Breiman, “Random forests,” Machine Learning, vol. 45, no. 1, pp. 5–32, 2001. [2] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, et al., “Scikit- learn: Machine learning in Python,” Journal of Machine Learning Research, vol. 12, pp. 2825–2830, 2011.

[3] Instagram Business, “Influencer marketing statistics and insights.” [Online]. Available: https://business.instagram.com/. [Accessed: Nov. 30, 2025].

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[5] I. Hassan, “Instagram Influencer Dataset,” 2025. [Data set]. Custom-generated for research purpose